Project Explanation

Step 1 – Introduction

- Start by providing a brief overview of the project, emphasizing its significance.
- Mention the problem or question the project aimed to address.
- Highlight the business context and objectives that led to the initiation of the project.

Step 2 - Data Collection

- Gather relevant data from various sources.
- This may involve obtaining data from databases, APIs, files, or other means.

Step 3 Data Cleaning and Preprocessing

- Clean and preprocess the raw data to handle missing values, outliers, and inconsistencies.
- This step is crucial for ensuring the quality of the data used for analysis.

Step 4 Exploratory Data Analysis (EDA)

- Perform exploratory data analysis to understand the characteristics of the data.
- Use visualizations, and summary metrics to identify patterns, trends, and outliers.

Step 5 Data Transformation

- Conduct data transformations and manipulations to create derived variables or features that enhance the analysis.
- This step may involve aggregations, merging datasets, or creating new variables to gain deeper insights.

Step 6 Model Deployment (if applicable)

- Describe the process of deploying the model in a production environment.

- Highlight the impact of the deployed model on business processes.

Step 7 Results and Impact

- Clearly communicate the results of your analysis and the impact on the business or problem at hand.
- Use quantitative metrics to demonstrate the success of the project.

Step 8 Challenges and Learnings

- Discuss any challenges you faced during the project and how you overcame them.
- Highlight key learnings and improvements for future projects.

Project Explanation – Google Image Scrapping

Introduction

- The primary objective of this project was to gain hands-on experience in web scraping using Python.
- I chose to focus on extracting images from Google Images as a practical application of data collection techniques.

Data Collection

- Data collection involved sending HTTP requests to Google Images using the 'requests' library in Python. The search query was set to 'MS Dhoni,' and the resulting HTML content was retrieved.

Libraries Used

1. urllib.request (urlopen)

Purpose

The urllib.request module is used for opening and reading URLs. In this project, urlopen is employed to fetch the HTML content of the Google Images search page.

2. BeautifulSoup

Purpose

BeautifulSoup is a Python library for pulling data out of HTML and XML files.

3. logging

The logging module in Python helps to record and manage messages or information about your program's execution, facilitating debugging and troubleshooting.

4. OS

The os module in Python provides functions to interact with the operating system, allowing tasks like creating directories.

EDA (Exploratory Data Analysis)

- Exploratory Data Analysis involved examining the structure of the HTML content, specifically the image tags retrieved.
- This step was crucial to understand the nature of the data and ensure its suitability for further analysis.

Data Transformation

- Data transformation mainly involved organizing the extracted images into a directory structure.
- The 'os' module was used to create a directory named 'images/' to store the scraped images.
- After importing the libraries, the directory was set up to save the scrapped images.
- Sending HTTP Request - An HTTP GET request is sent to the Google Images search page with a specific query (in this case, "MS Dhoni"). The response is stored in the response variable.

- HTML Parsing - BeautifulSoup is used to parse the HTML content obtained from the Google Images search page.
- Finding Image Tags for getting the scrapped images.

Result

- The project successfully retrieved a collection of images related to 'MS Dhoni' from Google Images.

Challenges and Learnings

- Challenges included handling potential blocks from Google due to scraping. To overcome this, I employed a 'User-Agent' header

Conclusion

- In conclusion, this image scraping project showcases the effective use of web scraping tools, combining BeautifulSoup and urllib, to retrieve and organize images from Google.